that has not grown deeply into nearby tissues. It has also not spread to the lymph nodes or other parts of the body.

Stage II and Stage III: These 2 stages indicate bigger tumors that have grown more deeply into the nearby tissues. They may have also spread to lymph nodes but not to other parts of the body.

Stage IV: This stage means that the cancer has spread to other parts of the body. It may also be called advanced or metastatic cancer.

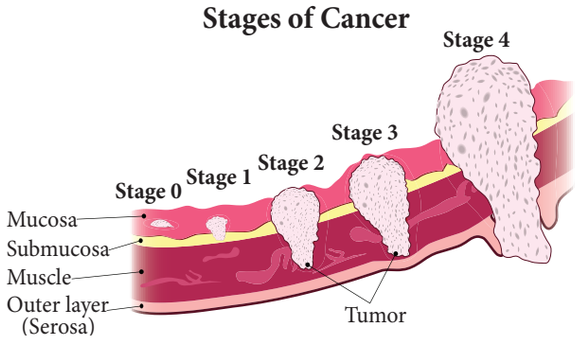


Fig 12.4 Stages of cancer

A classification system known as **TNM** is used to identify the size, the degree of growth and spread of the tumor.

- **T** stands for **"Tumor"** - it shows how large the primary tumor is and where it is located.
- **N** stands for **"Node"** - it shows whether the tumor has spread to the lymph nodes. If so, where and how many?
- **M** stands for **"Metastasis"** – it shows whether the cancer has spread to other parts of the body. If so, where and how much?



12.6 Diagnosis of cancer:

Detecting cancer is a multistage process which includes the following:

Imaging : Cancer detection often involves radiological imaging. Imaging involves creating pictures of areas inside the body that will help medical professionals to detect tumors. It is also used to check the spread of cancer and the progress of treatment. The most common imaging techniques used are

- Computed Tomography (CT)
- Magnetic Resonance Imaging (MRI)
- Positron Emission Tomography (PET)
- Ultra sound examination
- Endoscopic examinations
- Mammography and
- Isotopic diagnostics.

Tumor markers: These are biochemical substances that are produced by tumor cells or by other cells of the body in response to cancer. These substances can be found in the urine, blood, stool, tumor tissue and body fluids of individuals with cancer. They are produced at much higher levels in cancerous conditions. A tumor marker may be used along with other procedures to diagnose cancer. A few examples of tumor markers are **CA-125** (in ovarian cancer), **CA 15-3** (in breast cancer), **CEA** (in colon cancer), and **PSA** (in prostate cancer).

Biopsy test: A confirmatory test for the diagnosis of cancer involves an examination of the suspected tissue under a microscope by a pathologist. This is called a **"biopsy."** A Biopsy is an analysis of tissue removed from a living body to find out the presence, cause, or extent of cancer.



A pathologist is one who interprets and diagnoses the changes caused by disease in tissues and body fluids.

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